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Limit cycle oscillations in Goland wing with actuator free play

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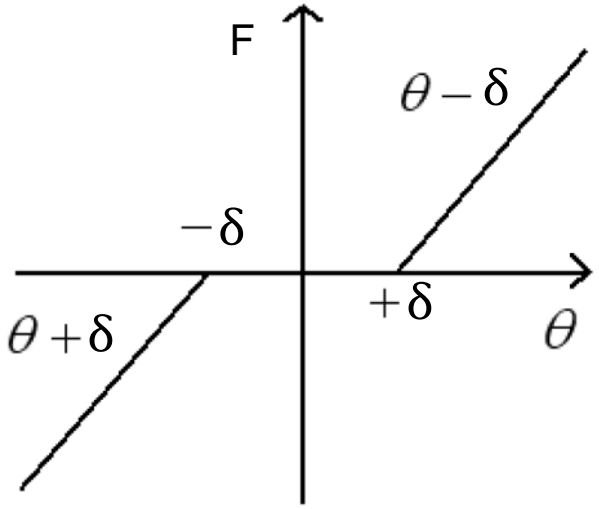
Introduction

1.1 Introduction
Free Play

- Aileron actuator stiffness K_β is null in the interval [-δ, +δ]. The resulting non-linearity leads to phenomena that cannot be studied with a linear model. In particular, the resulting hardening causes **Limit Cycle Oscillations.**

- Quasi-linearization is used to simplify the non-linearity while **retaining** the **dependency of the input-output relationship** on some characteristics of the input signal.

- The result of quasi-linearization is a (generally) complex gain that depends on the characteristics of the input signal.

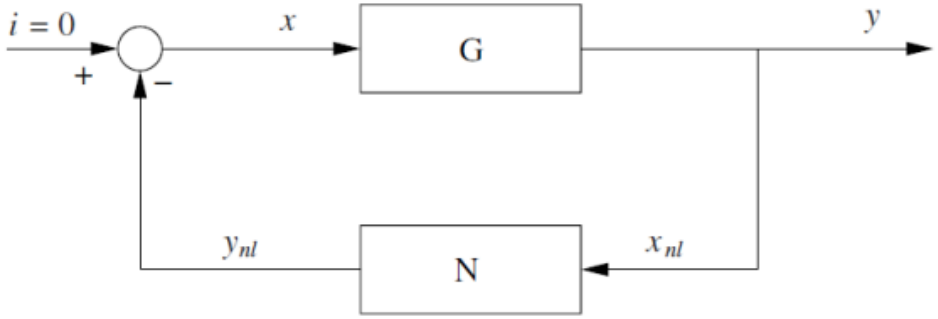


1.2 Introduction
Describing function

- Historically [1], it has been observed that the presence of **free-play** in the aileron results in **LCO** of the wing at certain speeds. A LCO is a constant amplitude oscillation in a system with dissipative forces.

- Quasi-linearization of the free-play was done using the **sinusoidal input describing function (SIDF)**. This describing function is well suited to analyze LCOs in systems with odd non-linearities [2].

- The combination of the DF and the system dynamics could be seen as this feedback system:



1.2 Introduction
Describing function - 2

- The **SIDF neglects all higher harmonics**. Therefore, the approximation will only be good if the system filters out these harmonics.

- The **SIDF** is computed with the following **formula** [2]:

$$N(U,\omega) = \frac{1}{U\pi}\int_0^{2\pi} f(U\sin\psi\,, U\omega\cos\psi)\mathrm{j}\,e^{-j\psi}d\psi$$

1.3 Introduction
Describing function - 3

- For the free-play problem, the **analytical form** of the describing function $\boldsymbol{N(A)}$ is given by:

$$N(A) = 0, \quad A \leq \delta$$

$$N(A) = \frac{2(-\mathrm{K}_\beta)}{\pi}\left[\sin^{-1}\left(\frac{\delta}{A}\right) + \frac{\delta}{A}\sqrt{1-\left(\frac{\delta}{A}\right)^2}\right] + \mathrm{K}_\beta, \quad A > \delta$$

- Since the free-play non-linearity is odd and single-valued the DF has no quadrature component. As a consequence, it **will not affect the phase** of the signal.

Resolution Methodology

2.1. Resolution Methodology
Goland Wing base problem modifications

- The LCO in this system will be characterized by the vibration of all the components of the state with a constant amplitude. The nature of this oscillation will be defined by the dynamics of the feedback system. In the frequency domain:

$$(-\omega^2\widehat{M} - j\omega U\hat{C}_{\text{theo}} + \widehat{K} + \widehat{K}_{nl}\text{N(A)} - U^2\widehat{\text{K}}_{\text{theo}})\tilde{q} = 0$$

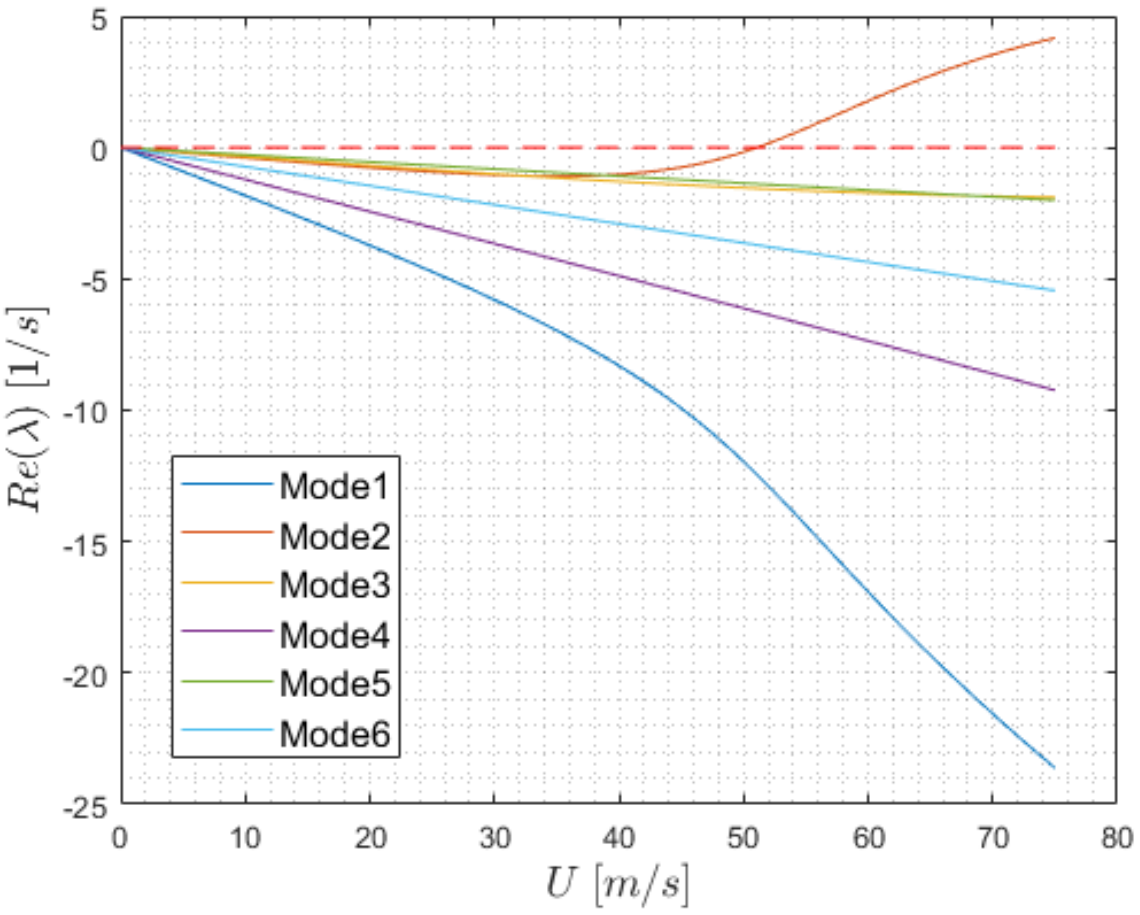
- Where:

1. $\widehat{K}$, $\widehat{M}$, $\hat{C}_{aer}$ and $\widehat{\text{K}}_{aer}$ are the **modal projections** of the **stiffness**, **mass**, **aerodynamic damping and stiffness** matrices, respectively. Theodorsen aerodynamics were used
2. $\boldsymbol{\omega}$ is the **frequency** of the LCO and $\boldsymbol{U}$ is the **airspeed**.
3. $\tilde{\boldsymbol{q}}$ is a vector that defines the **shape of the LCO in modal coordinates**.
4. $\widehat{K}_{nl}$ is derived from the PVW as $\widehat{K}_{nl} = C_{nl}^T C_{nl}$, where C_{nl} is a row vector containing the flap angle (β) components of the modes used in the modal projection.

2.1. Resolution Methodology
Goland Wing base problem modifications

- By the definition of the problem, the case of $N(A) = 0$ corresponds with the flutter point of the **Goland wing with zero actuator stiffness**. Consequently, this flutter point can be used as a **starting point** for identifications of LCO with larger amplitudes.

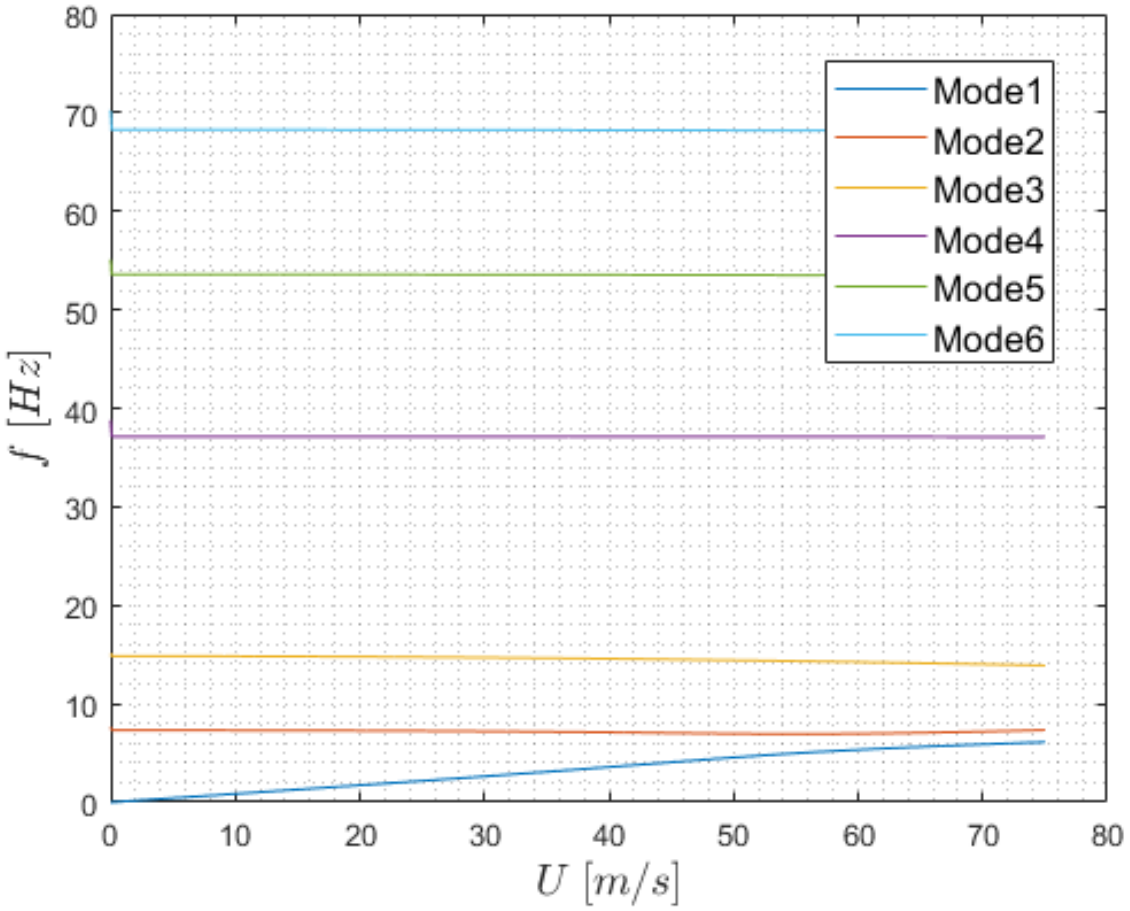
- The starting flutter point is at 51 m/s, with a frequency of 43 rad/s.



2.1. Resolution Methodology
Goland Wing base problem modifications

- By observing the damping and frequency plots of this model it can be concluded that the **flutter mechanism involves the first 2 modes**.

- With this in mind, the model for LCO identification is initially based on these 2 modes.



2.1. Resolution Methodology
Goland Wing equations development

- As previously presented, the problem is handled in the **frequency domain**.

- Since the model resembles a damped oscillator, complex eigenvectors are expected. For simplicity in the numerical treatment, they are **separated in the real and imaginary** part.

- With unknowns being **frequency** $\boldsymbol{\omega}$, **amplitude** $\boldsymbol{A}$, **velocity** $\boldsymbol{U}$ and **eigenvector** $\tilde{\boldsymbol{q}}$ (2x1), 2 more equations are needed for the problem to be solvable by fixing one of the variables and evaluating successively. They were chosen such that the $\boldsymbol{\beta}$ **motion associated to the LCO eigenvector corresponds to an oscillation of amplitude A and zero phase**.

$$\widehat{M}\ddot{\hat{q}} + \hat{C}\dot{\hat{q}} + \widehat{K}\hat{q} + \widehat{K}_{nl}\hat{q} = 0$$

$$-\omega^2\widehat{M}\tilde{q} + j\omega\hat{C}\tilde{q} + \widehat{K}\tilde{q} + \widehat{K}_{nl}\tilde{q} = 0$$

$$\left(-\omega^2\widehat{M} + \widehat{K} + \widehat{K}_{nl}\right)Re(\tilde{q}) - \omega\hat{C}Im(\tilde{q}) = 0$$

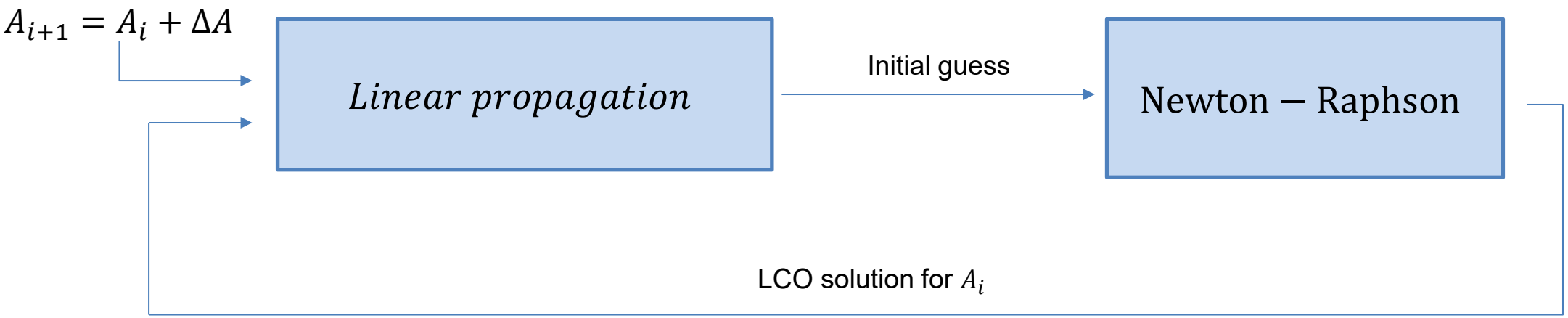
$$\left(-\omega^2\widehat{M} + \widehat{K} + \widehat{K}_{nl}\right)Im(\tilde{q}) + \omega\hat{C}Re(\tilde{q}) = 0$$

$$C_{nl}q_r = \mathrm{A}$$

$$C_{nl}q_{\mathrm{i}} = 0$$

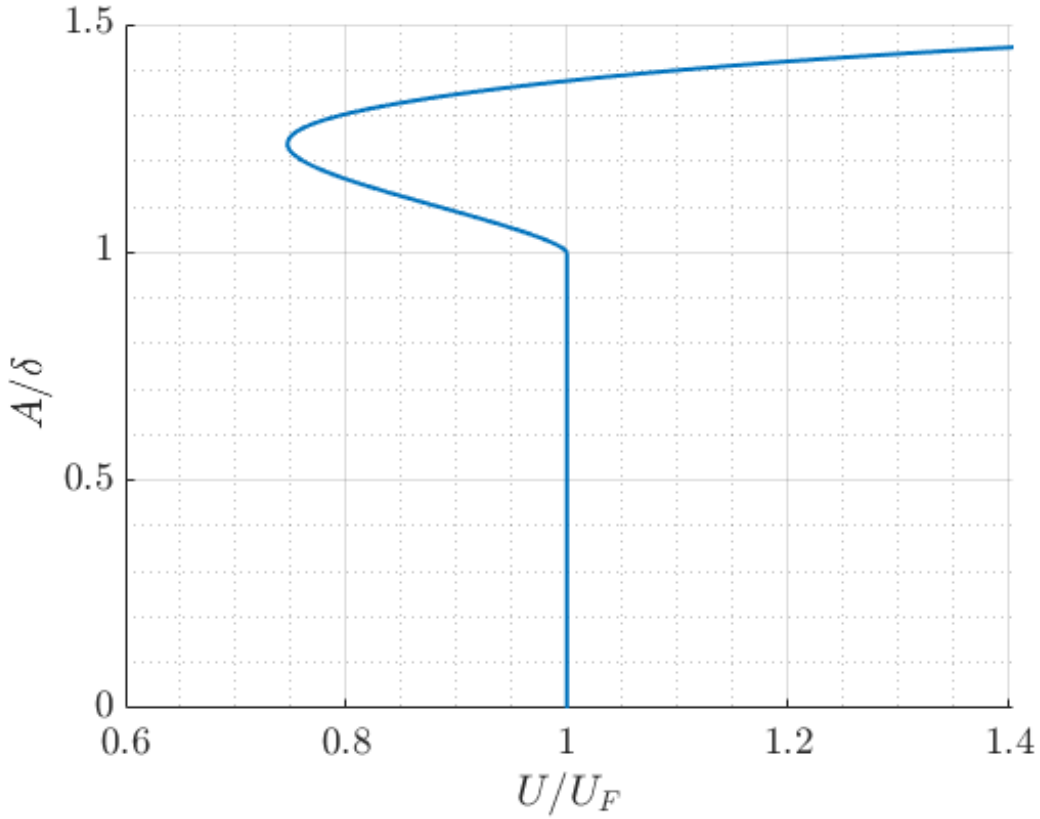
2.2. Resolution Methodology
LCO Computation

- The characteristics of the LCO as a function of A have been computed with a **continuation algorithm**. Starting from the flutter point, an initial guess is obtained by linear propagation. This guess is refined with a **Newton-Raphson** method up to a specified tolerance. The process is repeated over the desired range of amplitude.



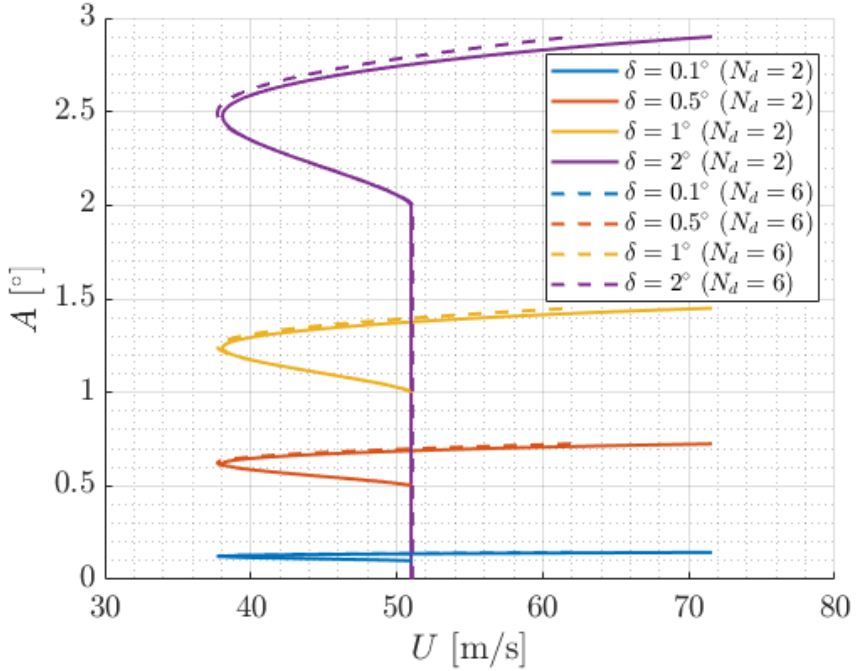
RESULTS

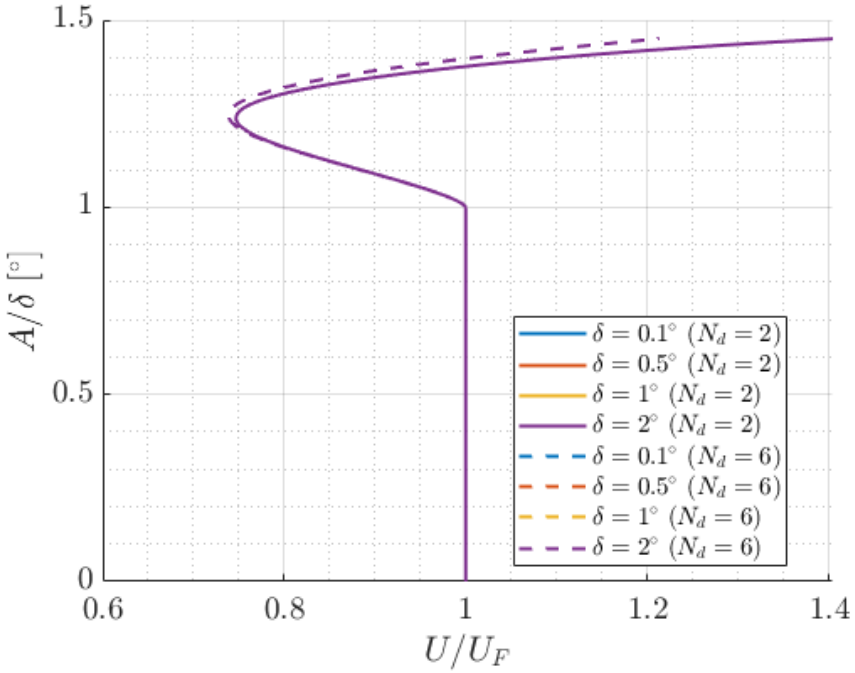
3.1. Results
LCO Curve



- The Goland wing model has a **subcritical LCO profile.** A **sufficiently large amplitude** perturbation can lead to an LCO at **velocities below U_F**.

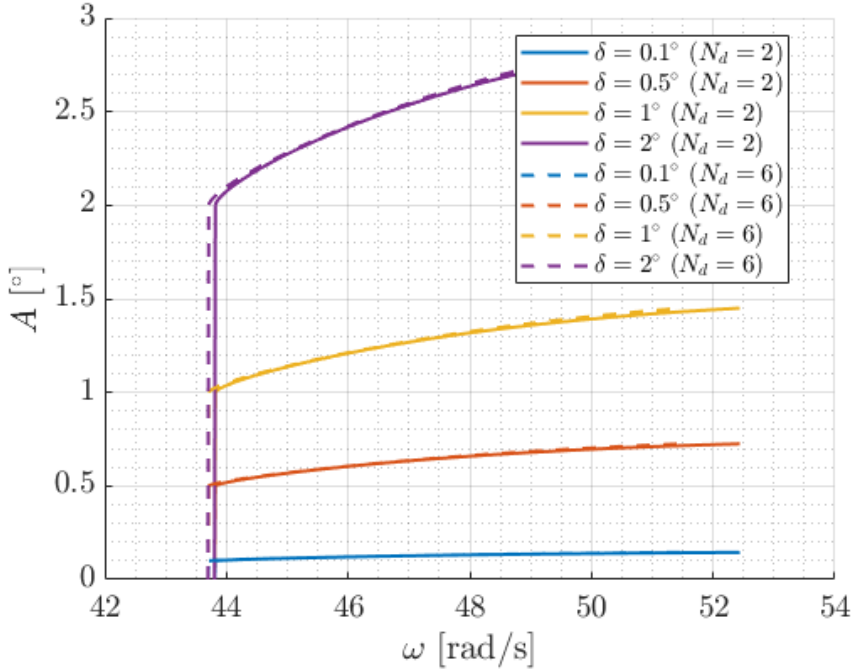
3.1. Results
LCO Curve - Amplitude

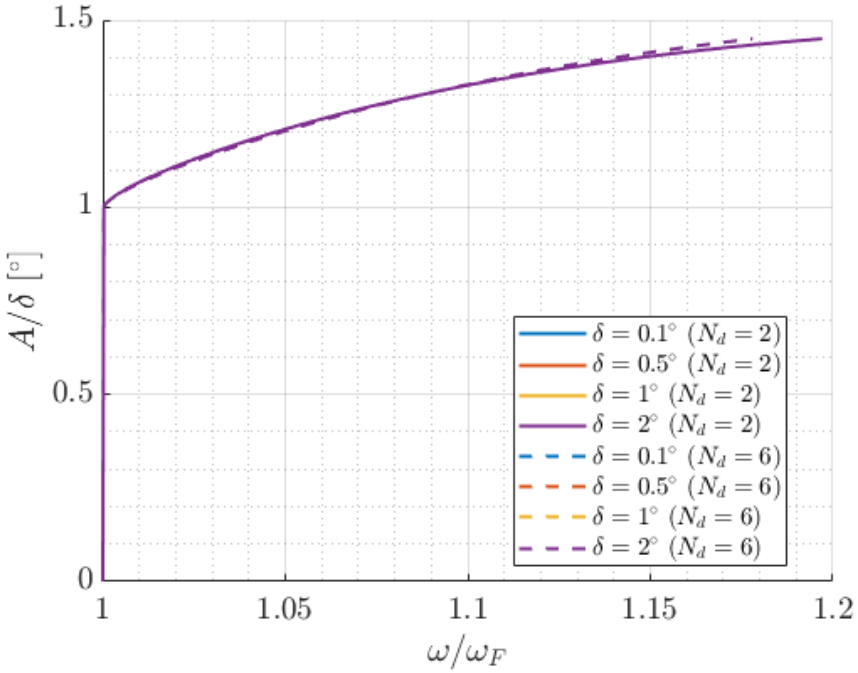




- Results considering 2 and 6 modes are practically **identical**, confirming that the LCO mechanism **involves mainly the first 2 modes**.
- The right figure demonstrates that the phenomenon **only depends on the A/δ ratio**. The response collapses into a single curve given by the non-linear ratio A/δ, as expected, since the DF is only a function of this ratio.

3.1. Results
LCO Curve - Frequency





- The LCO frequency increases by approx. **10%** as amplitude grows. This confirms the **hardening stiffness** behavior of the wing.

- Comments made for amplitude apply also to frequency plots.

3.2. Results
LCO Stability analysis - Methodology

- A study of the stability of the LCO is fundamental to understanding its effects on the performance of the system. In what follows, a static stability argument is presented.

- First, the equations are **linearized at constant velocity** around the **identified LCO**, considering perturbations in **amplitude, frequency, damping** ($\delta\sigma$) **and shape**.

$$\begin{pmatrix} \left(-2M\omega_0 - Ub\frac{dK_{theo}}{dk}\right)q_{r_0} + \left(UC_{theo,0} + Ub\omega_0\frac{dC_{theo}}{dk}\right)q_{i_0} & (UC_{theo,0}q_{r_0} - 2M\omega_0 q_{i_0}) & K_{nl}\frac{dN}{dA}(A_0)q_{r_0} & -M\omega_0^2 + K + K_{nl}N(A_0) - U^2K_{theo,0} & UC_{theo,0}\omega_0 \\ \left(-UC_{theo,0} - b\omega_0\frac{dC_{theo}}{dk}\right)q_{r_0} + \left(-2M\omega_0 - Ub\frac{dK_{theo}}{dk}\right)q_{i_0} & (-UC_{theo,0}q_{i_0} + 2M\omega_0 q_{r_0}) & K_{nl}\frac{dN}{dA}(A_0)q_{i_0} & -U\omega_0 C_{theo,0} & -M\omega_0^2 + K + K_{nl}N(A_0) - U^2K_{theo,0} \\ 0 & 0 & -1 & C_{nl} & 0 \\ 0 & 0 & 0 & 0 & C_{nl} \end{pmatrix} \begin{pmatrix} \delta\omega \\ \delta\sigma \\ \delta A \\ \delta q_r \\ \delta q_i \end{pmatrix} = \bar{0}$$

*Where the 0 subindex means the identified LCO at airspeed U.

3.2. Results
LCO Stability analysis – Methodology 2

- Treating $\boldsymbol{\delta A}$ as a known quantity, the relative variation of all other parameters can be derived by solving this linear system:

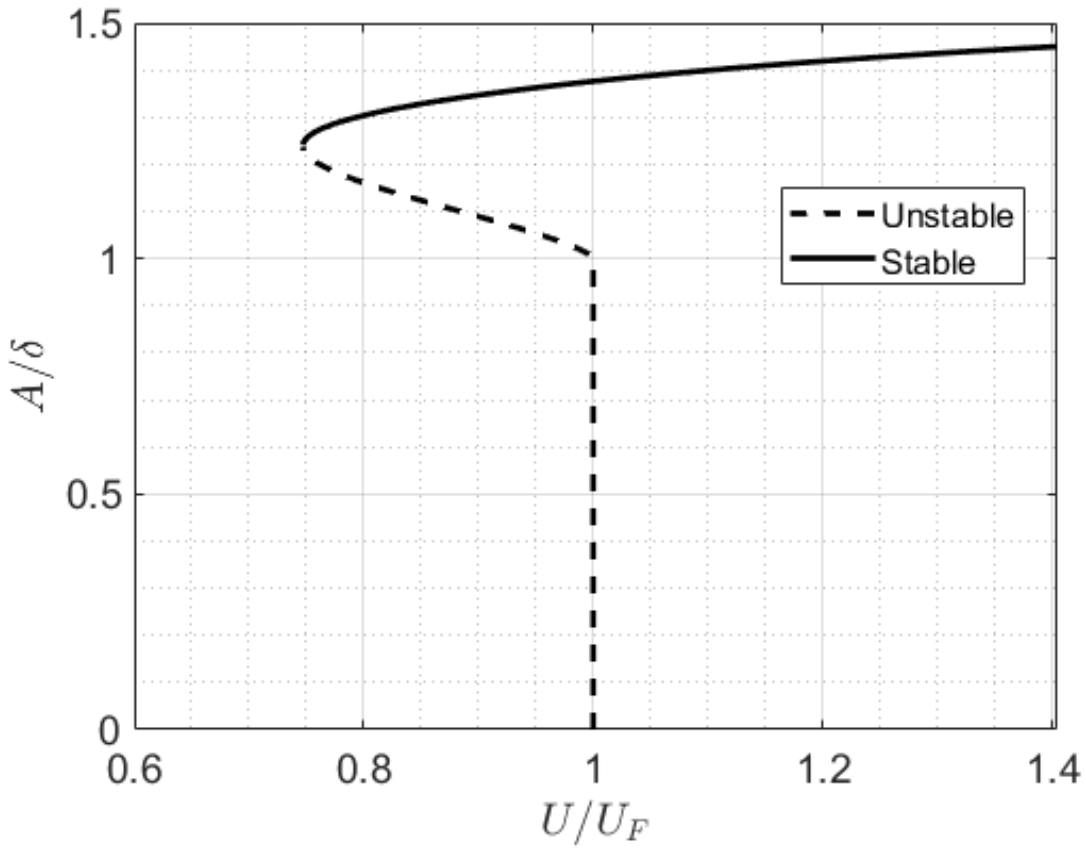
$$\begin{pmatrix} \left(-2M\omega_0 - Ub\frac{dK_{theo}}{dk}\right)q_{r_0} + \left(\mathrm{U}C_{theo,0} + Ub\omega_0\frac{dC_{theo}}{dk}\right)q_{i_0} & \left(UC_{theo,0}q_{r_0} - 2M\omega_0 q_{i_0}\right) & -M\omega_0^2 + K + K_{nl}N(A_0) - U^2K_{theo,0} & UC_{theo,0}\omega_0 \\ \left(-\mathrm{U}C_{theo,0} - b\omega_0\frac{dC_{theo}}{dk}\right)q_{r_0} + \left(-2M\omega_0 - Ub\frac{dK_{theo}}{dk}\right)q_{i_0} & \left(-UC_{theo,0}q_{i_0} + 2M\omega_0 q_{r_0}\right) & -\mathrm{U}\omega_0 C_{theo,0} & -M\omega_0^2 + K + K_{nl}N(A_0) - U^2K_{theo,0} \\ 0 & 0 & C_{nl} & 0 \\ 0 & 0 & 0 & C_{nl} \end{pmatrix} \begin{pmatrix} \frac{\delta\omega}{\delta A} \\ \frac{\delta\sigma}{\delta A} \\ \frac{\delta q_r}{\delta A} \\ \frac{\delta q_i}{\delta A} \end{pmatrix} = \begin{pmatrix} -K_{nl}\frac{dN}{dA}(A_0)q_{r_0} \\ -K_{nl}\frac{dN}{dA}(A_0)q_{i_0} \\ 1 \\ 0 \end{pmatrix}$$

- For the system to be statically stable the variation in damping ratio must oppose the variation in amplitude. In other words:

$$\frac{\delta\sigma}{\delta A} < 0$$

3.2. Results
LCO Stability analysis - Results

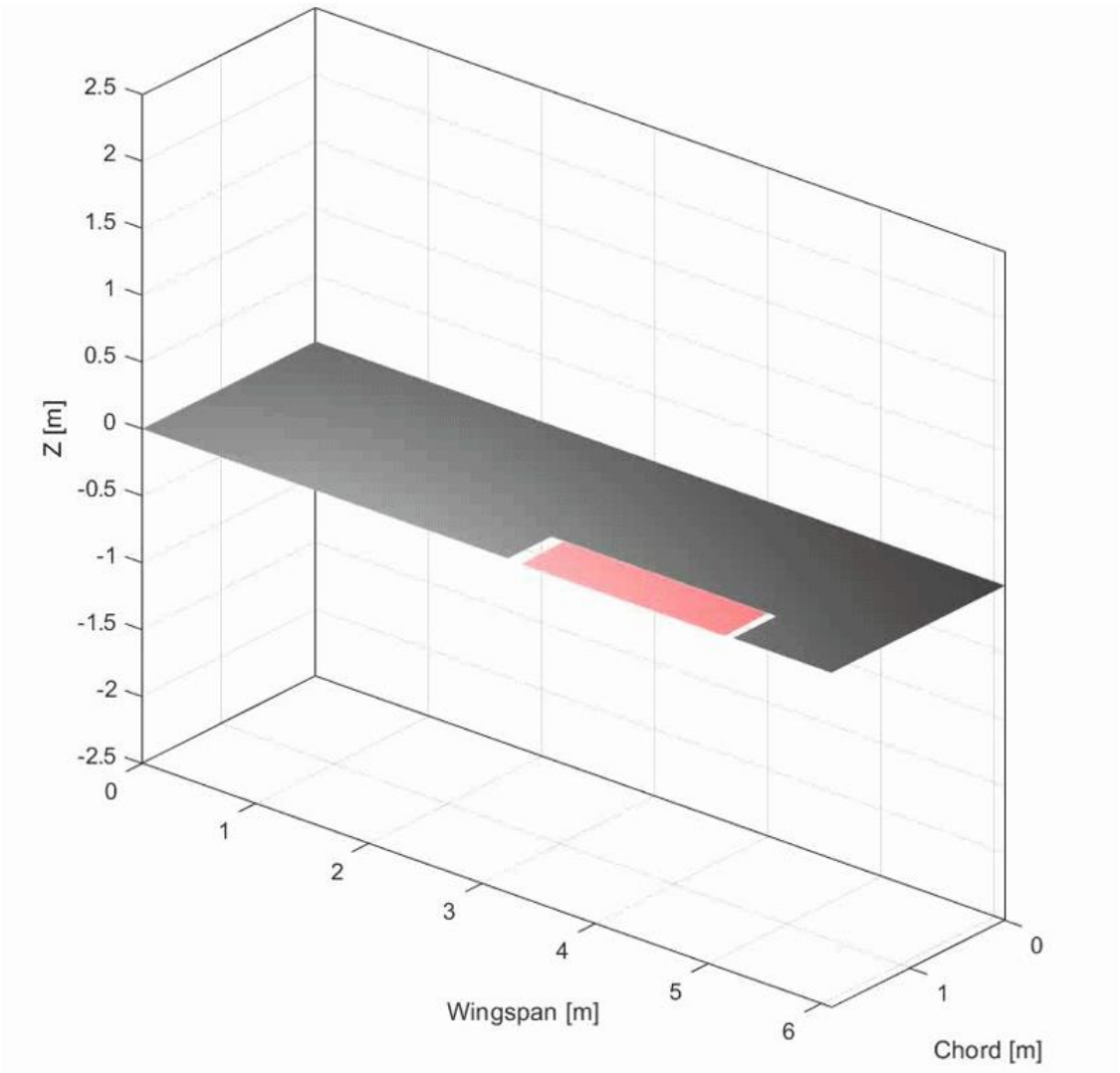
- LCO curve stability shows the expected behavior, where the **minimum velocity point** is the **first stable point**.
- **Low amplitude perturbations** lead to **unstable** LCOs. On the other hand, sufficiently **large perturbations** will lead to a **self-sustained** LCO.
- As in the previous cases, stability depends on the $\boldsymbol{A/\delta}$ **ratio**.



3.2. Results
LCO Visualization

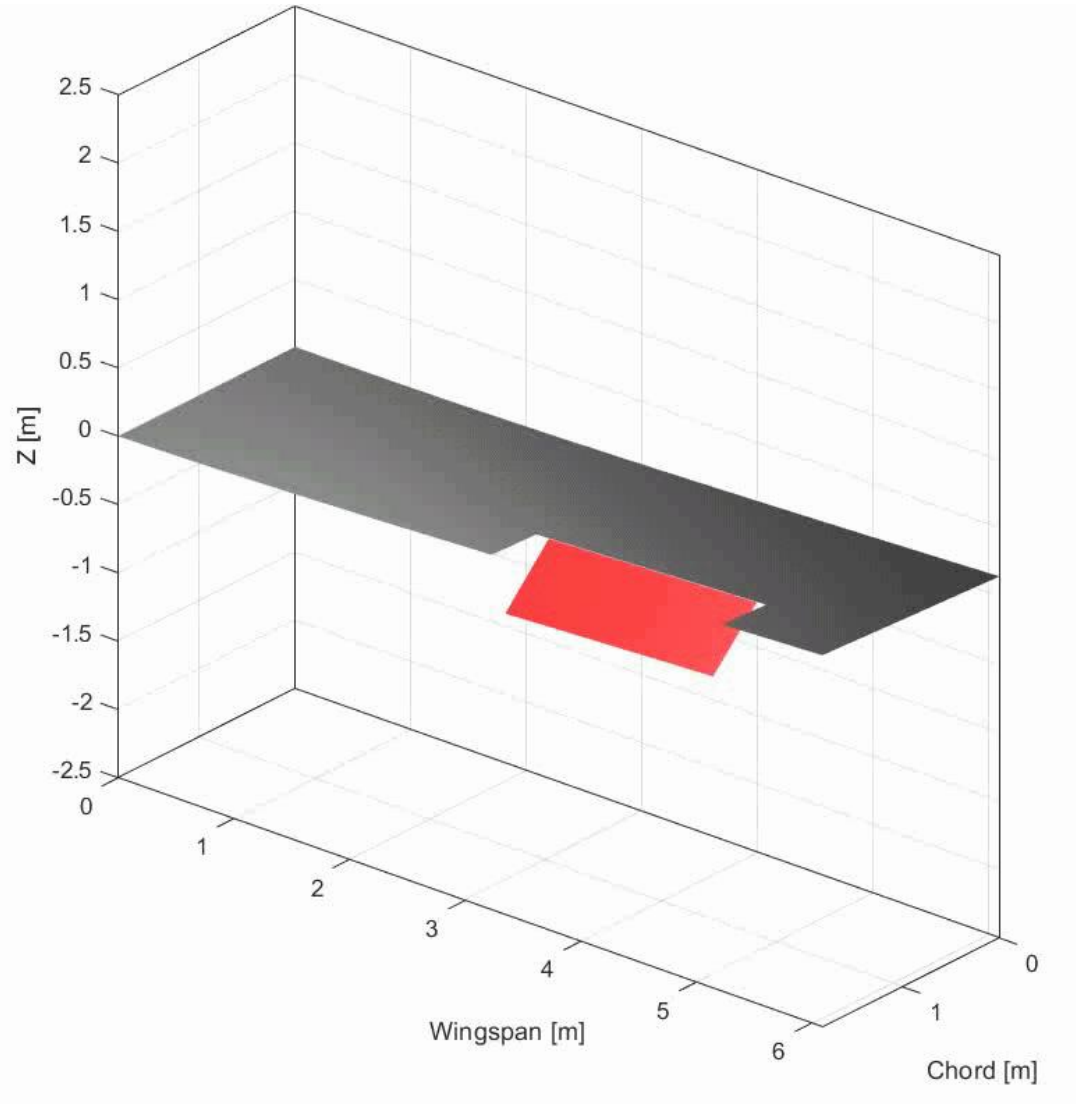
- This animation shows the shape of the LCO, at the lowest speed at which it is **stable**, according to previous results.

- **1 deg free play** in the actuator.



3.2. Results
LCO Visualization - Scaled

- Scaling the previous animation by a factor of 10 aids visualization.



Conclusions

4.1. Conclusions

- The **behavior** of the Goland Wing with free-play in the actuator identified with the DF is **subcritical**.
- A **dynamic stability analysis** would be required to gain a deeper understanding of the **stability and attractiveness of the LCOs**, as well as the transient behavior of the system.
- Results show actuator free-play should be considered in the design phase of a wing. In this case, **aeroelastic phenomena** are **relevant** starting **at 40 m/s**, while the flutter velocity of the **nominal system** without free-play is at **117 m/s**.
- Using the computed shape of the LCO, it is possible to compute its **effects on aircraft performance**, **lift** generation and **fatigue** life of the structure.

Sources

- [1] Conner, Mark D., et al. "Nonlinear behavior of a typical airfoil section with control surface freeplay: a numerical and experimental study." Journal of Fluids and structures 11.1 (1997): 89-109

- [2] A. Gelb and W. E. Vander Velde, "Multiple-Input Describing Functions and Nonlinear System Design." New York, NY, USA: McGraw-Hill, 1968.

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Q&A